Model Predictive Control

Lecture 7: Model Predictive Control

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02619 Model Predictive Control

Linear System and Linear Model

System, $\mathcal{S}$:

$$\begin{aligned}\boldsymbol{x}_{k+1}^s &= A_s\boldsymbol{x}_k^s + B_s u_k + G_s\boldsymbol{w}_k^s \\ \boldsymbol{y}_k &= C_{y,s}\boldsymbol{x}_k^s + \boldsymbol{v}_k^s \\ \boldsymbol{z}_k^s &= C_s\boldsymbol{x}_k^s\end{aligned}$$

$$\begin{bmatrix}\boldsymbol{w}_k^s \\ \boldsymbol{v}_k^s\end{bmatrix} \sim N_{iid}\left(\begin{bmatrix}0 \\ 0\end{bmatrix}, \begin{bmatrix}R_{ww}^s & R_{wv}^s \\ R_{vw}^s & R_{vv}^s\end{bmatrix}\right)$$

Model, $\mathcal{M}$:

$$\begin{aligned}\boldsymbol{x}_{k+1} &= A\boldsymbol{x}_k + Bu_k + G\boldsymbol{w}_k \\ \boldsymbol{y}_k &= C_y\boldsymbol{x}_k + \boldsymbol{v}_k \\ \boldsymbol{z}_k &= C\boldsymbol{x}_k\end{aligned}$$

$$\begin{bmatrix}\boldsymbol{w}_k \\ \boldsymbol{v}_k\end{bmatrix} \sim N_{iid}\left(\begin{bmatrix}0 \\ 0\end{bmatrix}, \begin{bmatrix}R_{ww} & R_{wv} \\ R_{vw} & R_{vv}\end{bmatrix}\right)$$

Filtering

Stationary Kalman Filter - Design

Discrete Algebraic Riccati Equation

$$P = APA' + GR_{ww}G' - (APC' + GR_{wv})(CPC' + R_{vv})^{-1}(APC' + GR_{wv})'$$

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P = dare(A',C',G*Rww*G',Rvv,G*Rwv)
```

- Note that $P = 0$ is a solution for state space models in innovation form. This solution is not necessarily found using the `dare` routine.
- In such situation it may be found by solving the discrete algebraic Riccati equation by iteration.

Filter Gain Computations

$$R_e = CPC' + R_{vv}$$
$$K_{fx} = PC'R_e^{-1}$$
$$K_{fw} = R_{wv}R_e^{-1}$$

Stationary Kalman Filter - Compute

Stationary Filter

Input: $\hat{x}_{k-1|k-1}, \hat{w}_{k-1|k-1}, u_{k-1|k} = u_{k-1}, y_k$

1. $\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + B\hat{u}_{k-1|k} + G\hat{w}_{k-1|k-1}$
2. $\hat{y}_{k|k-1} = C_y\hat{x}_{k|k-1}$
3. $e_k = y_k - \hat{y}_{k|k-1}$
4. $\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_{fx}e_k$
5. $\hat{w}_{k|k} = K_{fw}e_k$

Output: $\hat{x}_{k|k}, \hat{w}_{k|k}$

Stationary Kalman Filter - Prediction

The Kalman predictions are

$$\begin{aligned}
\hat{x}_{k+1|k} &= A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
\hat{x}_{k+j+1|k} &= A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
\hat{z}_{k+j|k} &= C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N
\end{aligned}$$

The Markov parameters (impulse response parameters)

$$H_i = CA^{i-1}B$$

may be used to obtain the following for the Kalman predictions ($N = 4$)

$$\underbrace{\begin{bmatrix} \hat{z}_{k+1|k} \\ \hat{z}_{k+2|k} \\ \hat{z}_{k+3|k} \\ \hat{z}_{k+4|k} \end{bmatrix}}_{Z_k} = \overbrace{\underbrace{\begin{bmatrix} CA \\ CA^2 \\ CA^3 \\ CA^4 \end{bmatrix}}_{\Phi_x} \hat{x}_{k|k} + \underbrace{\begin{bmatrix} CG \\ CAG \\ CA^2G \\ CA^3G \end{bmatrix}}_{\Phi_w} \hat{w}_{k|k}}^{b_k} + \underbrace{\begin{bmatrix} H_1 & 0 & 0 & 0 \\ H_2 & H_1 & 0 & 0 \\ H_3 & H_2 & H_1 & 0 \\ H_4 & H_3 & H_2 & H_1 \end{bmatrix}}_{\Gamma} \underbrace{\begin{bmatrix} \hat{u}_{k|k} \\ \hat{u}_{k+1|k} \\ \hat{u}_{k+2|k} \\ \hat{u}_{k+3|k} \end{bmatrix}}_{U_k}$$

$$Z_k = b_k + \Gamma U_k \qquad b_k = \Phi_x \hat{x}_{k|k} + \Phi_w \hat{w}_{k|k}$$

Regulation

Regulator

Object Function Terms

$$\phi_z = \frac{1}{2}\sum_{j=1}^{N} \left\| W_z(\hat{z}_{k+j|k} - \hat{r}_{k+j|k}) \right\|_2^2$$

$$\phi_u = \frac{1}{2}\sum_{j=0}^{N-1} \left\| W_u(\hat{u}_{k+j|k} - \bar{u}_{k+j|k}) \right\|_2^2$$

$$\phi_{\Delta u} = \frac{1}{2}\sum_{j=0}^{N-1} \left\| W_{\Delta u}\Delta\hat{u}_{k+j|k} \right\|_2^2$$

Predictions

$$\begin{aligned}
\hat{x}_{k+1|k} &= A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
\hat{x}_{k+j+1|k} &= A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
\hat{z}_{k+j|k} &= C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N
\end{aligned}$$

Input Constraints

Input Constraints

$$\hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} \qquad j = 0, 1, \ldots, N-1$$
$$\Delta u_{\min} \leq \Delta \hat{u}_{k+j|k} \leq \Delta u_{\max} \qquad j = 0, 1, \ldots, N-1$$

Soft Output Constraints

Soft Output Constraints - Lower Soft Output Constraint

$$\hat{z}_{k+j|k} \geq \hat{r}_{\min,k+j|k} - \hat{s}_{k+j|k} \qquad j = 1, 2, \ldots, N$$

$$\hat{s}_{k+j|k} \geq 0 \qquad j = 1, 2, \ldots, N$$

$$\phi_s = \frac{1}{2} \sum_{j=1}^{N} \left\| W_{s,2} \hat{s}_{k+j|k} \right\|_2^2 + \sum_{j=1}^{N} \left\| W_{s,1} \hat{s}_{k+j|k} \right\|_1$$

Soft Output Constraints - Upper Soft Output Constraint

$$\hat{z}_{k+j|k} \leq \hat{r}_{\max,k+j|k} + \hat{t}_{k+j|k} \qquad j = 1, 2, \ldots, N$$

$$\hat{t}_{k+j|k} \geq 0 \qquad j = 1, 2, \ldots, N$$

$$\phi_t = \frac{1}{2} \sum_{j=1}^{N} \left\| W_{t,2} \hat{t}_{k+j|k} \right\|_2^2 + \sum_{j=1}^{N} \left\| W_{t,1} \hat{t}_{k+j|k} \right\|_1$$

Stacked Vectors

$N = 4$

$$Z_k = \begin{bmatrix} \hat{z}_{k+1|k} \\ \hat{z}_{k+2|k} \\ \hat{z}_{k+3|k} \\ \hat{z}_{k+4|k} \end{bmatrix} \qquad U_k = \begin{bmatrix} \hat{u}_{k|k} \\ \hat{u}_{k+1|k} \\ \hat{u}_{k+2|k} \\ \hat{u}_{k+3|k} \end{bmatrix} \qquad \Delta U_k = \begin{bmatrix} \Delta\hat{u}_{k|k} \\ \Delta\hat{u}_{k+1|k} \\ \Delta\hat{u}_{k+2|k} \\ \Delta\hat{u}_{k+3|k} \end{bmatrix}$$

$$R_k = \begin{bmatrix} \hat{r}_{k+1|k} \\ \hat{r}_{k+2|k} \\ \hat{r}_{k+3|k} \\ \hat{r}_{k+4|k} \end{bmatrix} \qquad R_{\min,k} = \begin{bmatrix} \hat{r}_{\min,k+1|k} \\ \hat{r}_{\min,k+2|k} \\ \hat{r}_{\min,k+3|k} \\ \hat{r}_{\min,k+4|k} \end{bmatrix} \qquad R_{\max,k} = \begin{bmatrix} \hat{r}_{\max,k+1|k} \\ \hat{r}_{\max,k+2|k} \\ \hat{r}_{\max,k+3|k} \\ \hat{r}_{\max,k+4|k} \end{bmatrix}$$

$$S_k = \begin{bmatrix} \hat{s}_{k+1|k} \\ \hat{s}_{k+2|k} \\ \hat{s}_{k+3|k} \\ \hat{s}_{k+4|k} \end{bmatrix} \qquad T_k = \begin{bmatrix} \hat{t}_{k+1|k} \\ \hat{t}_{k+2|k} \\ \hat{t}_{k+3|k} \\ \hat{t}_{k+4|k} \end{bmatrix}$$

Prediction

The Kalman predictions are

$$\begin{aligned}
\hat{x}_{k+1|k} &= A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
\hat{x}_{k+j+1|k} &= A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
\hat{z}_{k+j|k} &= C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N
\end{aligned}$$

The Markov parameters (impulse response parameters)

$$H_i = CA^{i-1}B \qquad i = 1, 2, \ldots, N$$

may be used to obtain the following for the Kalman predictions ($N = 4$)

$$\underbrace{\begin{bmatrix} \hat{z}_{k+1|k} \\ \hat{z}_{k+2|k} \\ \hat{z}_{k+3|k} \\ \hat{z}_{k+4|k} \end{bmatrix}}_{Z_k} = \overbrace{\underbrace{\begin{bmatrix} CA \\ CA^2 \\ CA^3 \\ CA^4 \end{bmatrix}}_{\Phi_x} \hat{x}_{k|k} + \underbrace{\begin{bmatrix} CG \\ CAG \\ CA^2G \\ CA^3G \end{bmatrix}}_{\Phi_w} \hat{w}_{k|k}}^{b_k} + \underbrace{\begin{bmatrix} H_1 & 0 & 0 & 0 \\ H_2 & H_1 & 0 & 0 \\ H_3 & H_2 & H_1 & 0 \\ H_4 & H_3 & H_2 & H_1 \end{bmatrix}}_{\Gamma} \underbrace{\begin{bmatrix} \hat{u}_{k|k} \\ \hat{u}_{k+1|k} \\ \hat{u}_{k+2|k} \\ \hat{u}_{k+3|k} \end{bmatrix}}_{U_k}$$

$$Z_k = b_k + \Gamma U_k \qquad b_k = \Phi_x \hat{x}_{k|k} + \Phi_w \hat{w}_{k|k}$$

Objective Term: ϕ_z

$$
\begin{aligned}
\phi_z &= \frac{1}{2}\sum_{j=1}^{N}\left\|W_z(\hat{z}_{k+j|k}-\hat{r}_{k+j|k})\right\|_2^2 = \frac{1}{2}\left\|\bar{W}_z(Z_k-R_k)\right\|_2^2 \\
&= \frac{1}{2}\left\|\bar{W}_z(b_k+\Gamma U_k-R_k)\right\|_2^2 = \frac{1}{2}\left\|\bar{W}_z(\Gamma U_k-c_k)\right\|_2^2 \\
&= \frac{1}{2}U_k'\left[\Gamma'\bar{W}_z'\bar{W}_z\Gamma\right]U_k + \left[-\Gamma'\bar{W}_z'\bar{W}_z c_k\right]'U_k + \left[\frac{1}{2}c_k'\bar{W}_z'\bar{W}_z c_k\right] \\
&= \frac{1}{2}U_k'\left[(\bar{W}_z\Gamma)'(\bar{W}_z\Gamma)\right]U_k + \left[-(\bar{W}_z\Gamma)'\bar{W}_z c_k\right]'U_k + \left[\frac{1}{2}c_k'\bar{W}_z'\bar{W}_z c_k\right] \\
&= \frac{1}{2}U_k'H_zU_k + g_z'U_k + \rho_z
\end{aligned}
$$

$$
\begin{aligned}
&\bar{W}_z = I_N \otimes W_z \\
&c_k = R_k - b_k \\
&H_z = (\bar{W}_z\Gamma)'(\bar{W}_z\Gamma) \\
&g_z = -(\bar{W}_z\Gamma)'\bar{W}_z c_k = M_z c_k \qquad M_z = -(\bar{W}_z\Gamma)'\bar{W}_z \\
&\rho_z = \frac{1}{2}c_k'\bar{W}_z'\bar{W}_z c_k
\end{aligned}
$$

Objective Term: ϕ_u

$$
\begin{aligned}
\phi_u &= \frac{1}{2}\sum_{j=0}^{N-1}\left\|W_u(\hat{u}_{k+j|k}-\bar{u}_{k+j|k})\right\|_2^2 = \frac{1}{2}\left\|\bar{W}_u(U_k-\bar{U}_k)\right\|_2^2 \\
&= \frac{1}{2}U_k'\left[\bar{W}_u'\bar{W}_u\right]U_k + \left[-\bar{W}_u'\bar{W}_u\bar{U}_k\right]'U_k + \left[\frac{1}{2}\bar{U}_k'\bar{W}_u'\bar{W}_u\bar{U}_k\right] \\
&= \frac{1}{2}U_k'H_uU_k + g_u'U_k + \rho_u
\end{aligned}
$$

$$
\begin{aligned}
\bar{W}_u &= I_N \otimes W_u \\
H_u &= \bar{W}_u'\bar{W}_u \\
g_u &= -\bar{W}_u'\bar{W}_u\bar{U}_k = M_u\bar{U}_k \qquad M_u = -\bar{W}_u'\bar{W}_u = -H_u \\
\rho_u &= \frac{1}{2}\bar{U}_k'\bar{W}_u'\bar{W}_u\bar{U}_k
\end{aligned}
$$

Objective Term: $\phi_{\Delta u}$

$$
\begin{aligned}
\phi_{\Delta u} &= \frac{1}{2}\sum_{j=0}^{N-1}\left\|W_{\Delta u}\Delta u_{k+j|k}\right\|_2^2 = \frac{1}{2}\left\|\bar{W}_{\Delta u}\Delta U_k\right\|_2^2 = \frac{1}{2}\left\|\bar{W}_{\Delta u}(\Lambda U_k - I_0\hat{u}_{k-1|k})\right\|_2^2 \\
&= \frac{1}{2}U_k'\left[\Lambda'\bar{W}_{\Delta u}'\bar{W}_{\Delta u}\Lambda\right]U_k + \left[-\Lambda'\bar{W}_{\Delta u}'\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k}\right]'U_k \\
&\quad + \left[\frac{1}{2}\hat{u}_{k-1|k}I_0'\bar{W}_{\Delta u}'\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k}\right] \\
&= \frac{1}{2}U_k'\left[(\bar{W}_{\Delta u}\Lambda)'(\bar{W}_{\Delta u}\Lambda)\right]U_k + \left[-(\bar{W}_{\Delta u}\Lambda)'\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k}\right]'U_k \\
&\quad + \left[\frac{1}{2}(\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k})'(\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k})\right] \\
&= \frac{1}{2}U_k'H_{\Delta u}U_k + g_{\Delta u}'U_k + \rho_{\Delta u}
\end{aligned}
$$

$$
\begin{aligned}
&\bar{W}_{\Delta u} = I_N \otimes W_{\Delta u} \\
&H_{\Delta u} = (\bar{W}_{\Delta u}\Lambda)'(\bar{W}_{\Delta u}\Lambda) \\
&g_{\Delta u} = -(\bar{W}_{\Delta u}\Lambda)'\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k} = M_{\Delta u}\hat{u}_{k-1|k} \qquad M_{\Delta u} = -(\bar{W}_{\Delta u}\Lambda)'\bar{W}_{\Delta u}I_0 \\
&\rho_{\Delta u} = \frac{1}{2}(\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k})'(\bar{W}_{\Delta u}I_0\hat{u}_{k-1|k})
\end{aligned}
$$

Objective Function: $\phi = \phi_z + \phi_u + \phi_{\Delta u}$

$$
\begin{aligned}
\phi &= \phi_z + \phi_u + \phi_{\Delta u} \\
&= \frac{1}{2} U_k' H_z U_k + g_z' U_k + \rho_z \\
&\quad + \frac{1}{2} U_k' H_u U_k + g_u' U_k + \rho_u \\
&\quad + \frac{1}{2} U_k' H_{\Delta u} U_k + g_{\Delta u}' U_k + \rho_{\Delta u} \\
&= \frac{1}{2} U_k' \left[H_z + H_u + H_{\Delta u} \right] U_k + \left[g_z + g_u + g_{\Delta u} \right]' U_k + \left[\rho_z + \rho_u + \rho_{\Delta u} \right] \\
&= \frac{1}{2} U_k' H U_k + g' U_k + \rho
\end{aligned}
$$

$$
\begin{aligned}
H &= H_z + H_u + H_{\Delta u} \\
g &= g_z + g_u + g_{\Delta u} \\
\rho &= \rho_z + \rho_u + \rho_{\Delta u}
\end{aligned}
$$

Input Bound Constraints

The input bound constraints

$$\hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} \qquad j = 0, 1, \ldots, N-1$$

can be expressed as $(N = 4)$

$$\underbrace{\begin{bmatrix} \hat{u}_{\min,k|k} \\ \hat{u}_{\min,k+1|k} \\ \hat{u}_{\min,k+2|k} \\ \hat{u}_{\min,k+3|k} \end{bmatrix}}_{U_{\min,k}} \leq \underbrace{\begin{bmatrix} \hat{u}_{k|k} \\ \hat{u}_{k+1|k} \\ \hat{u}_{k+2|k} \\ \hat{u}_{k+3|k} \end{bmatrix}}_{U_k} \leq \underbrace{\begin{bmatrix} \hat{u}_{\max,k|k} \\ \hat{u}_{\max,k+1|k} \\ \hat{u}_{\max,k+2|k} \\ \hat{u}_{\max,k+3|k} \end{bmatrix}}_{U_{\max,k}}$$

$$U_{\min,k} \leq U_k \leq U_{\max,k}$$

Input Rate-of-Movement

The input rate-of-movement for $N = 4$

$$\Delta U_k = \begin{bmatrix} \Delta\hat{u}_{k|k} \\ \Delta\hat{u}_{k+1|k} \\ \Delta\hat{u}_{k+2|k} \\ \Delta\hat{u}_{k+3|k} \end{bmatrix} = \begin{bmatrix} \hat{u}_{k|k} - \hat{u}_{k-1|k} \\ \hat{u}_{k+1|k} - \hat{u}_{k|k} \\ \hat{u}_{k+2|k} - \hat{u}_{k+1|k} \\ \hat{u}_{k+3|k} - \hat{u}_{k+2|k} \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} I & 0 & 0 & 0 \\ -I & I & 0 & 0 \\ 0 & -I & I & 0 \\ 0 & 0 & -I & I \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} \hat{u}_{k|k} \\ \hat{u}_{k+1|k} \\ \hat{u}_{k+2|k} \\ \hat{u}_{k+3|k} \end{bmatrix}}_{U_k} - \underbrace{\begin{bmatrix} I \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{I_0} \hat{u}_{k-1|k}$$

can be expressed as

$$\Delta U_k = \Lambda U_k - I_0 \hat{u}_{k-1|k}$$

Input Rate-of-Movement Constraint

The input rate-of-movement constraints

$$\Delta u_{\min} \leq \Delta \hat{u}_{k+j|k} \leq \Delta u_{\max} \qquad j = 0, 1, \ldots, N-1$$

can be expressed as $(N = 4)$

$$\underbrace{\begin{bmatrix} \Delta u_{\min} \\ \Delta u_{\min} \\ \Delta u_{\min} \\ \Delta u_{\min} \end{bmatrix}}_{\Delta U_{\min}} \leq \underbrace{\begin{bmatrix} I & 0 & 0 & 0 \\ -I & I & 0 & 0 \\ 0 & -I & I & 0 \\ 0 & 0 & -I & I \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} \hat{u}_{k|k} \\ \hat{u}_{k+1|k} \\ \hat{u}_{k+2|k} \\ \hat{u}_{k+3|k} \end{bmatrix}}_{U_k} - \underbrace{\begin{bmatrix} I \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{I_0} \hat{u}_{k-1|k} \leq \underbrace{\begin{bmatrix} \Delta u_{\max} \\ \Delta u_{\max} \\ \Delta u_{\max} \\ \Delta u_{\max} \end{bmatrix}}_{\Delta U_{\max}}$$

that are equivalent to the constraints

$$\Delta U_{\min} + I_0 \hat{u}_{k-1|k} \leq \Lambda U_k \leq \Delta U_{\max} + I_0 \hat{u}_{k-1|k}$$

Soft Output Constraint - Lower Bound

The soft output constraints

$$\begin{aligned} \hat{z}_{k+j|k} &\geq \hat{r}_{\min,k+j|k} - \hat{s}_{k+j|k} \qquad && j = 1, 2, \ldots, N \\ \hat{s}_{k+j|k} &\geq 0 && j = 1, 2, \ldots, N \end{aligned}$$

can be expressed as

$$\begin{aligned} Z_k &\geq R_{\min,k} - S_k \\ S_k &\geq 0 \end{aligned}$$

that using $Z_k = \Gamma U_k + b_k$ can be transformed to

$$Z_k = \Gamma U_k + b_k \geq R_{\min,k} - S_k \quad \Leftrightarrow \quad \Gamma U_k + S_k \geq R_{\min,k} - b_k$$

Consequently, the soft output constraints are

$$\begin{aligned} \Gamma U_k + S_k &\geq R_{\min,k} - b_k \\ S_k &\geq 0 \end{aligned}$$

Soft Output Constraints - Lower Bound

Using that $\hat{s}_{k+j|k} \geq 0$ for $j = 1, 2, \ldots, N$, the objective function term related to the lower bound constraints can be expressed as

$$\begin{aligned}\phi_s &= \frac{1}{2}\sum_{j=1}^{N}\left\|W_{s,2}\hat{s}_{k+j|k}\right\|_2^2 + \sum_{j=1}^{N}\left\|W_{s,1}\hat{s}_{k+j|k}\right\|_1 = \frac{1}{2}\left\|\bar{W}_{s,2}S_k\right\|_2^2 + \left\|\bar{W}_{s,1}S_k\right\|_1 \\ &= \frac{1}{2}S_k'\left[\bar{W}_{s,2}'\bar{W}_{s,2}\right]S_k + \left[\bar{W}_{s,1}e\right]'S_k \\ &= \frac{1}{2}S_k'H_sS_k + g_s'S_k\end{aligned}$$

with

$$\begin{aligned}\bar{W}_{s,2} &= I_N \otimes W_{s,2} \\ \bar{W}_{s,1} &= I_N \otimes W_{s,1} \\ H_s &= \bar{W}_{s,2}'\bar{W}_{s,2} \\ g_s &= \bar{W}_{s,1}e \qquad e = \begin{bmatrix}1 & 1 & \ldots & 1\end{bmatrix}'\end{aligned}$$

Soft Output Constraint - Upper Bound

The soft output constraints

$$\begin{aligned} \hat{z}_{k+j|k} &\leq \hat{r}_{\max,k+j|k} + \hat{t}_{k+j|k} \qquad && j = 1, 2, \ldots, N \\ \hat{t}_{k+j|k} &\geq 0 && j = 1, 2, \ldots, N \end{aligned}$$

can be expressed as

$$\begin{aligned} Z_k &\leq R_{\max,k} + T_k \\ T_k &\geq 0 \end{aligned}$$

that using $Z_k = \Gamma U_k + b_k$ can be transformed to

$$Z_k = \Gamma U_k + b_k \leq R_{\max,k} + T_k \quad \Leftrightarrow \quad \Gamma U_k + T_k \leq R_{\max,k} - b_k$$

Consequently, the soft output constraints are

$$\begin{aligned} \Gamma U_k - T_k &\leq R_{\max,k} - b_k \\ T_k &\geq 0 \end{aligned}$$

Soft Output Constraints - Upper Bound

Using that $\hat{t}_{k+j|k} \geq 0$ for $j = 1, 2, \ldots, N$, the objective function term related to the lower bound constraints can be expressed as

$$\begin{aligned}
\phi_s &= \frac{1}{2}\sum_{j=1}^{N}\left\|W_{t,2}\hat{t}_{k+j|k}\right\|_2^2 + \sum_{j=1}^{N}\left\|W_{t,1}\hat{t}_{k+j|k}\right\|_1 = \frac{1}{2}\left\|\bar{W}_{t,2}T_k\right\|_2^2 + \left\|\bar{W}_{t,1}T_k\right\|_1 \\
&= \frac{1}{2}T_k'\left[\bar{W}_{t,2}'\bar{W}_{t,2}\right]T_k + \left[\bar{W}_{t,1}e\right]'T_k \\
&= \frac{1}{2}T_k'H_tT_k + g_t'T_k
\end{aligned}$$

with

$$\begin{aligned}
\bar{W}_{t,2} &= I_N \otimes W_{t,2} \\
\bar{W}_{t,1} &= I_N \otimes W_{t,1} \\
H_s &= \bar{W}_{t,2}'\bar{W}_{t,2} \\
g_s &= \bar{W}_{t,1}e \qquad e = \begin{bmatrix}1 & 1 & \ldots & 1\end{bmatrix}'
\end{aligned}$$

Unconstrained Regulator

$$
\begin{aligned}
\min_{u} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N
\end{aligned}
$$

Unconstrained Regulator

$$
\begin{aligned}
\min_{u} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N
\end{aligned}
$$

$$
\min_{U_k} \quad \phi = \frac{1}{2} U_k' H U_k + g' U_k + \rho
$$

$$
\begin{aligned}
b_k &= \Phi_x \hat{x}_{k|k} + \Phi_w \hat{w}_{k|k} \\
c_k &= R_k - b_k \\
g &= g_z + g_u + g_{\Delta u} = M_z c_k + M_u \bar{U}_k + M_{\Delta u} \hat{u}_{k-1|k}
\end{aligned}
$$

Input Bound Constrained Regulator

$$
\begin{aligned}
\min_{u,s,t} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} && j = 0, 1, \ldots, N-1
\end{aligned}
$$

Input Bound Constrained Regulator

$$
\begin{aligned}
\min_{u,s,t} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} && j = 0, 1, \ldots, N-1
\end{aligned}
$$

$$
\begin{aligned}
\min_{U_k} \quad & \phi = \frac{1}{2} U_k' H U_k + g' U_k + \rho \\
s.t. \quad & U_{\min,k} \leq U_k \leq U_{\max,k}
\end{aligned}
$$

Input Constrained Regulator

$$
\begin{aligned}
\min_{u} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} && j = 0, 1, \ldots, N-1 \\
& \Delta u_{\min} \leq \Delta\hat{u}_{k+j|k} \leq \Delta u_{\max} && j = 0, 1, \ldots, N-1
\end{aligned}
$$

Input Constrained Regulator

$$
\begin{aligned}
\min_{u} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} && j = 0, 1, \ldots, N-1 \\
& \Delta u_{\min} \leq \Delta\hat{u}_{k+j|k} \leq \Delta u_{\max} && j = 0, 1, \ldots, N-1
\end{aligned}
$$

$$
\begin{aligned}
\min_{U_k} \quad & \phi = \frac{1}{2} U_k' H U_k + g' U_k + \rho \\
s.t. \quad & U_{\min,k} \leq U_k \leq U_{\max,k} \\
& \Delta U_{\min} + I_0 \hat{u}_{k-1|k} \leq \Lambda U_k \leq \Delta U_{\max} + I_0 \hat{u}_{k-1|k}
\end{aligned}
$$

Input Constrained Regulator with Soft Output Constraints

$$
\begin{aligned}
\min_{u,s,t} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} + \phi_s + \phi_t \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} && j = 0, 1, \ldots, N-1 \\
& \Delta u_{\min} \leq \Delta\hat{u}_{k+j|k} \leq \Delta u_{\max} && j = 0, 1, \ldots, N-1 \\
& \hat{z}_{k+j|k} \geq \hat{r}_{\min,k+j|k} - \hat{s}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{s}_{k+j|k} \geq 0 && j = 1, 2, \ldots, N \\
& \hat{z}_{k+j|k} \leq \hat{r}_{\max,k+j|k} + \hat{t}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{t}_{k+j|k} \geq 0 && j = 1, 2, \ldots, N
\end{aligned}
$$

$$
\begin{aligned}
\min_{u,s,t} \quad & \phi = \phi_z + \phi_u + \phi_{\Delta u} + \phi_s + \phi_t \\
s.t. \quad & \hat{x}_{k+1|k} = A\hat{x}_{k|k} + B\hat{u}_{k|k} + G\hat{w}_{k|k} \\
& \hat{x}_{k+j+1|k} = A\hat{x}_{k+j|k} + B\hat{u}_{k+j|k} && j = 1, 2, \ldots, N-1 \\
& \hat{z}_{k+j|k} = C\hat{x}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{u}_{\min,k+j|k} \leq \hat{u}_{k+j|k} \leq \hat{u}_{\max,k+j|k} && j = 0, 1, \ldots, N-1 \\
& \Delta u_{\min} \leq \Delta\hat{u}_{k+j|k} \leq \Delta u_{\max} && j = 0, 1, \ldots, N-1 \\
& \hat{z}_{k+j|k} \geq \hat{r}_{\min,k+j|k} - \hat{s}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{s}_{k+j|k} \geq 0 && j = 1, 2, \ldots, N \\
& \hat{z}_{k+j|k} \leq \hat{r}_{\max,k+j|k} + \hat{t}_{k+j|k} && j = 1, 2, \ldots, N \\
& \hat{t}_{k+j|k} \geq 0 && j = 1, 2, \ldots, N
\end{aligned}
$$

$$
\begin{aligned}
\min_{U_k,S_k,T_k} \quad & \phi = \frac{1}{2}\begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix}' \begin{bmatrix} H & 0 & 0 \\ 0 & H_s & 0 \\ 0 & 0 & H_t \end{bmatrix} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} + \begin{bmatrix} g \\ g_s \\ g_t \end{bmatrix}' \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} + \rho \\
s.t. \quad & \begin{bmatrix} U_{\min,k} \\ 0 \\ 0 \end{bmatrix} \leq \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \leq \begin{bmatrix} U_{\max,k} \\ \infty \\ \infty \end{bmatrix} \\
& \begin{bmatrix} \Delta U_{\min} + I_0\hat{u}_{k-1|k} \\ R_{\min,k} - b_k \\ -\infty \end{bmatrix} \leq \begin{bmatrix} \Lambda & 0 & 0 \\ \Gamma & I & 0 \\ \Gamma & 0 & -I \end{bmatrix} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \leq \begin{bmatrix} \Delta U_{\max} + I_0\hat{u}_{k-1|k} \\ \infty \\ R_{\max,k} - b_k \end{bmatrix}
\end{aligned}
$$

$$\min_{U_k, S_k, T_k} \quad \phi = \frac{1}{2} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix}' \begin{bmatrix} H & 0 & 0 \\ 0 & H_s & 0 \\ 0 & 0 & H_t \end{bmatrix} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} + \begin{bmatrix} g \\ g_s \\ g_t \end{bmatrix}' \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} + \rho$$

$$s.t. \qquad \begin{bmatrix} U_{\min,k} \\ 0 \\ 0 \end{bmatrix} \leq \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \leq \begin{bmatrix} U_{\max,k} \\ \infty \\ \infty \end{bmatrix}$$

$$\begin{bmatrix} \Delta U_{\min} + I_0 \hat{u}_{k-1|k} \\ R_{\min,k} - b_k \\ -\infty \end{bmatrix} \leq \begin{bmatrix} \Lambda & 0 & 0 \\ \Gamma & I & 0 \\ \Gamma & 0 & -I \end{bmatrix} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \leq \begin{bmatrix} \Delta U_{\max} + I_0 \hat{u}_{k-1|k} \\ \infty \\ R_{\max,k} - b_k \end{bmatrix}$$

$$\begin{aligned} b_k &= \Phi_x \hat{x}_{k|k} + \Phi_w \hat{w}_{k|k} \\ c_k &= R_k - b_k \\ g &= M_z c_k + M_u \bar{U}_k + M_{\Delta u} \hat{u}_{k-1|k} \end{aligned}$$

$$
\begin{aligned}
\min_{U_k, S_k, T_k} \quad & \phi = \frac{1}{2} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix}' \begin{bmatrix} H & 0 & 0 \\ 0 & H_s & 0 \\ 0 & 0 & H_t \end{bmatrix} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} + \begin{bmatrix} g \\ g_s \\ g_t \end{bmatrix}' \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} + \rho \\
s.t. \quad & \begin{bmatrix} U_{\min,k} \\ 0 \\ 0 \end{bmatrix} \leq \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \leq \begin{bmatrix} U_{\max,k} \\ \infty \\ \infty \end{bmatrix} \\
& \begin{bmatrix} \Delta U_{\min} + I_0 \hat{u}_{k-1|k} \\ R_{\min,k} - b_k \\ -\infty \end{bmatrix} \leq \begin{bmatrix} \Lambda & 0 & 0 \\ \Gamma & I & 0 \\ \Gamma & 0 & -I \end{bmatrix} \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \leq \begin{bmatrix} \Delta U_{\max} + I_0 \hat{u}_{k-1|k} \\ \infty \\ R_{\max,k} - b_k \end{bmatrix}
\end{aligned}
$$

is equivalent to

$$
\begin{aligned}
\min_{\bar{x}} \quad & \phi = \frac{1}{2} \bar{x}' \bar{H} \bar{x} + \bar{g}' \bar{x} + \bar{\rho} \\
s.t. \quad & \bar{l} \leq \bar{x} \leq \bar{u} \\
& \bar{b}_l \leq \bar{A} \bar{x} \leq \bar{b}_u
\end{aligned}
$$

with

$$
\bar{x} = \begin{bmatrix} U_k \\ S_k \\ T_k \end{bmatrix} \quad \bar{H} = \begin{bmatrix} H & 0 & 0 \\ 0 & H_s & 0 \\ 0 & 0 & H_t \end{bmatrix} \quad \bar{g} = \begin{bmatrix} g \\ g_s \\ g_t \end{bmatrix} \quad \bar{l} = \begin{bmatrix} U_{\min,k} \\ 0 \\ 0 \end{bmatrix} \quad \bar{u} = \begin{bmatrix} U_{\max,k} \\ \infty \\ \infty \end{bmatrix}
$$

$$
\bar{A} = \begin{bmatrix} \Lambda & 0 & 0 \\ \Gamma & I & 0 \\ \Gamma & 0 & -I \end{bmatrix} \quad \bar{b}_l = \begin{bmatrix} \Delta U_{\min} + I_0 \hat{u}_{k-1|k} \\ R_{\min,k} - b_k \\ -\infty \end{bmatrix} \quad \bar{b}_u = \begin{bmatrix} \Delta U_{\max} + I_0 \hat{u}_{k-1|k} \\ \infty \\ R_{\max,k} - b_k \end{bmatrix}
$$

Implementation - MPC Compute

1. Convert from physical variables to deviation variables
2. Kalman Filter Compute
3. Regulator Compute
4. Convert from deviation variables to physical variables

Questions and Comments

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